



Expert Recommendations — EdTech Island Platform

What I Would Improve If This Were My Codebase — Architecture, Security, Backend, UI/UX & Gemini AI Features

Scope: Architecture • Code Quality • Backend & DB • Security • UI/UX • 18 Gemini AI Features **Discipline:** Ponytail Zero-Waste Engineering

💡 **Note:** Professional recommendations based on a full deep-read of the codebase, database schema, dev logs, and Obsidian architecture notes. Prioritized strictly by technical and security impact.

🏗️ 1. ARCHITECTURE

● Move to a Single Monorepo Build System

CRITICAL

Problem: 4 separate Vite builds share zero code. supabase.js, ChatInterface.jsx, Card.jsx, Layout.jsx are copy-pasted 3–4 times across portals.

Fix: Extract packages/shared/ with npm workspaces. One change to supabase.js propagates everywhere automatically.

```
packages/
  shared/
    components/ ← Card, ChatInterface, Layout
    hooks/      ← usePresence, useAnnouncements
    lib/        ← supabase.js (single source of truth)
```

● Study Island Should Be a React App, Not 2,272-Line Vanilla JS

CRITICAL

Problem: script.js is a monolith. No modularity, no tree-shaking, cannot be tested. Adding any new chapter means editing one massive file.

Fix: Convert to Vite + React. Each chapter = a lazy-loaded route component. New chapters = new folders only.

```
/study-island/src/chapters/
  science/LightAndShadows/ ← lazy loaded
  math/SpatialGeometry/
  geo/EarthEngine/
```

● No API Layer — Supabase Queries Directly in 30+ Components

IMPORTANT

Business logic tangled with UI. Any table rename requires changes across 30+ files.

Fix: Create thin src/services/ layer. Components call classService.getAll(), never supabase.from(...) directly.

💻 2. CODE QUALITY

● Progress.jsx Is Entirely Static Fake Data

CRITICAL

Problem: chartDataByRange, study streak "10 Days", and subject mastery percentages are all hardcoded. Students are looking at fake grades and don't know it.

Fix: Connect to quiz_scores and student_test_attempts tables. Calculate real streak from last_active_date in profiles.

● Test Scores Stored in localStorage Only — Lost on Device Change

CRITICAL

Quiz results saved only to localStorage['student_test_results']. Cleared browser data = lost academic record.

Fix: After grading, INSERT into test_submissions(id, student_id, test_id, score, grade, answers_json, submitted_at). Keep localStorage as cache only.

● ToDoList.jsx Is Ephemeral — Resets on Every Refresh

IMPORTANT

25 lines of component. 0% Supabase integration. Teacher tasks disappear on navigation.

Fix: Wire to Supabase todos table scoped by teacher_id with optimistic updates.

● Class Dropdowns Hardcoded in 5+ Files + No Loading Skeletons

IMPORTANT

`mockClasses = ['Class 1st'...]` copy-pasted across `LiveClass`, `Inbox`, `Timetable`, and more. Components show raw white flashes while fetching.

Fix: Single `useClasses()` hook querying the DB once. Add shimmer skeleton CSS components on all data-fetching views.

🔧 3. BACKEND & DATABASE

● No Row-Level Security (RLS) — Any User Can Read All Data in DevTools

CRITICAL

All queries use the public anon key. Any student can query quiz_scores, messages, or profiles of every other student from the browser console.

Minimum policies needed:

```
CREATE POLICY "students_own_scores" ON quiz_scores
  USING (auth.uid() = student_id);

CREATE POLICY "message_visibility" ON messages
  USING (auth.uid() IN (
    SELECT user_id FROM conversation_participants
    WHERE conversation_id = messages.conversation_id
  ));
```

● No student_class_enrollment Table — Class 6 Students See Class 8 Tests

IMPORTANT

No table linking students to classes. Every student sees all tests from all classes.

Fix: Create student_enrollments(student_id, class_id, enrolled_at). Filter all live class & test queries by enrolled class.

● No DB Migration System + Schema Column Inconsistency

IMPORTANT

Schema changes are made directly in Supabase Studio — no history. announcements queries mix createdAt (camelCase) and created_at (snake_case) causing silent null returns.

Fix: Use Supabase CLI with supabase/migrations/ folder. Standardize on created_at (Postgres convention).

🔒 4. SECURITY

● Auth is Entirely Client-Side Trust — Students Can Impersonate Teachers

CRITICAL

Role checks (if user.role === 'teacher') happen in React state. Any student can open DevTools, set localStorage.edtech_teacher_user, and access teacher tools.

Fix: Every sensitive action validated server-side via Supabase auth.uid() RLS policies. Never trust client state for authorization.

● URL Injection / XSS Session Hijacking in Chats.jsx

CRITICAL

Chats.jsx reads ?user= from URL and writes it directly to localStorage. A crafted link can hijack any student's session role.

Fix: Remove URL parameter user parsing entirely. Session identity from Supabase Auth tokens only.

🐛 5. UI / UX

● No Mobile Responsiveness — Unusable on Phones & Tablets

CRITICAL

Fixed 240px sidebars, calc(100vh - 145px) heights. Students in India will primarily use phones. Current experience is broken on mobile.

Fix: Add responsive CSS breakpoints. Sidebar collapses to hamburger drawer below 768px.

● No Empty States, No WCAG Accessibility, No Typed Confirmations

IMPORTANT

Blank white areas on empty data. Icon buttons have no aria-label. Destructive actions ("Clear All Notices") have only browser confirm() dialogs.

Fix: Add illustrated empty state cards + CTA. Add WCAG 2.1 ARIA labels. Build styled confirmation modals with type-to-confirm for destructive actions.

🌟 6. MISSING CORE PLATFORM FEATURES

1. Student Attendance Tracking System

FEATURE

No attendance system exists. Teachers cannot mark present/absent. This is the #1 most-requested feature in school management software.

Teacher attendance grid → attendance Supabase table → Admin monthly reports → Parent push alerts.

2. Homework / Assignment Submission System

FEATURE

No concept of homework with deadlines. Students cannot submit files or text. Teachers cannot grade submitted work.

assignments(due_date, class_id) + assignment_submissions(student_id, file_url, text_answer, graded_score).

3. Gamification & XP System in Study Island

FEATURE

Study Island has the visual aesthetic of a game but no actual game loop. Students have zero incentive to return daily.

Earn XP: Chapter complete (+50), Quiz 90%+ (+100), 7-day streak (+200), All 5 modalities (+75). Class leaderboard in Student Dashboard. Proven to increase daily usage 40–60%.

4. Parent Portal (Read-Only) + Web Push Notifications

FEATURE

No parent visibility into child's progress. No push alerts when live classes start or tests are assigned.

Lightweight read-only portal scoped to one student record. Web Push API notifications (zero dependencies) for live class starts and test assignments.

🤖 7. GEMINI AI-POWERED FEATURES (18 RECOMMENDATIONS)

9. 📷 "Snap & Solve" Handwritten Homework & Diagram Doubt Solver

AI — VISION

Gemini 2.0 Flash (Vision + Text)

Target: Student Portal — Student photographs handwritten math problem, textbook page, or physics ray diagram. Gemini identifies the exact concept and provides Socratic step-by-step guidance (not just the answer). Automatically links to the matching Study Island 3D chapter.

10. 🔍 At-Risk Student Early Warning System

AI — ANALYTICS

Gemini 2.0 Flash (Text + Structured Data Analysis)

Target: Admin + Teacher Portals — Weekly AI-generated "Student Health Report" analyzing quiz score drops, attendance, and Study Island inactivity. Color-coded risk tiers: On Track / Needs Attention / Intervention Required. Catches failing students weeks before exams.

11. 📖 Personalized AI Learning Path Engine

AI — EMBEDDINGS

Gemini 2.0 Flash (Text + Embeddings)

Target: Study Island + Student Dashboard — Gemini analyzes quiz history and uses text embeddings to understand semantic knowledge gaps. If weak on "refraction," predicts weakness in "Snell's Law" even without being tested. Reorders Study Island chapters per student.

12. 🎙️ Live Class Transcript & AI Smart Notes Generator

AI — LONG CONTEXT

Gemini 2.0 Pro (2M Token Audio/Video Context)

Target: Teacher + Student LiveClass — Teacher uploads class recording after Meet ends. Gemini transcribes with speaker ID, extracts key concepts and formulas, generates structured student notes, and creates a mini-quiz pushed live. Students who missed class get a full AI-summarized brief.

13. 📄 CBSE Exam Prediction Engine

AI — LONG CONTEXT

Gemini 2.0 Pro (Long Context PDF)

Target: Teacher + Student Progress — Ingests 5+ years of CBSE past papers and cross-references with teacher's published chapters. Outputs: Top 5 most likely exam questions, chapter coverage gap analysis, and difficulty calibration vs. CBSE standards.

14. 🎨 Chapter Storybook Illustrator

AI — IMAGE GENERATION

Imagen 3

Target: Study Island Stories Tab — For history, geography, and science concepts where 3D is too expensive to build, Imagen 3 generates rich, curriculum-accurate storybook illustrations on demand. Generated once, saved to Cloudflare R2, reused at zero per-view cost.

15. 🎧 Canvas-Aware 3D Lab Voice Tutor (Real-Time)

AI — LIVE MULTIMODAL

Gemini Multimodal Live API (Real-Time Audio + Vision)

Target: Study Island 3D Chapters — Inside Shadow Lab, student activates microphone. Browser streams WebRTC audio + canvas screenshots (1fps) to Gemini Live. Gemini sees the 3D scene and talks back in real-time voice: *"I can see your light source is 10 units away — try the aperture slider!"* A 1-on-1 physics tutor inside the simulation for every student simultaneously.

16. 🌐 Vernacular Multilingual Voice Narration (Hindi / Hinglish)

AI — AUDIO TTS

Gemini Native Audio TTS + Translation

Target: Study Island + Stories tab — Toggle button on every chapter page. Students hear narrations in Natural Hindi, conversational Hinglish (most natural for Indian CBSE students), or English. Audio generated once, saved to Cloudflare R2 — zero per-play cost.

17. 🏆 Dynamic Personalized Achievement Badge Generator

AI — IMAGE GENERATION

Imagen 3

Target: Student Progress Achievements — On milestone unlock, Imagen 3 generates a unique personalized Cyber-Glass holographic badge with the student's name embedded in the visual aesthetic. Downloadable as a shareable social image. Stored to Cloudflare R2.

18. 🧡 AI Study Buddy Matchmaking

AI — STRUCTURED ANALYSIS

Gemini 2.0 Flash (Text + Structured Analysis)

Target: Student Portal — Study Groups — Analyzes quiz scores per subject and pairs complementary students (Student A: Strong Science/Weak Math with Student B: Strong Math/Weak Science). Opens dedicated private chat with topic suggestions. Peer learning outperforms solo study in CBSE contexts.

19. 🗣️ Voice-to-Structured MCQ Creator

AI — AUDIO + JSON

Gemini 2.0 Flash (Audio + Structured JSON Output)

Target: Teacher Question Bank — Teacher speaks: *"Question: What happens when light travels from denser to rarer medium at critical angle? Correct: total internal reflection. Wrong: refraction, absorption, dispersion."* Gemini converts to structured MCQ JSON inserted directly into question_bank. No typing required.

20. 📧 AI Parent Communication Draft Generator

AI — TEXT

Gemini 2.0 Flash (Text)

Target: Teacher Portal — Classes — Select a struggling student. One-click generates a professional, empathetic parent letter pre-filled with actual Supabase performance data: quiz average, last Study Island activity, missed assignments. Teacher edits and emails or sends via WhatsApp.

21. 💬 Chapter-Scoped RAG Expert Chat

AI — RAG

Gemini 2.0 Flash (Retrieval-Augmented Generation)

Target: Study Island — Chapter Sidebar — Each chapter has a dedicated chat that answers only from that chapter's approved curriculum content stored in Supabase vector embeddings. Curriculum-accurate, prevents exam shortcutting, teachers can inject approved answers.

22. 🛡️ AI Chat Moderation & Academic Integrity Guardian

AI — CLASSIFICATION

Gemini 2.0 Flash (Text Classification via Supabase Edge Function)

Target: All Portal Chat Systems — Real-time moderation of every message before storage. Flags bullying (alerts admin silently). Detects answer-sharing (blocks + notifies teacher). Identifies student distress signals (pings counselor).

23. 🖼️ 3D Scene Auto-Generator from Text Description

AI — CODE GENERATION

Gemini 2.0 Pro (Code + Three.js Config)

Target: SuperAdmin Curriculum Hub — Teacher types: *"Create a 3D scene showing the water cycle with evaporation, condensation, precipitation labels and click-to-learn zones."* Gemini generates a Three.js scene config JSON that Study Island loads directly. Simple scenes deployable in minutes.

24. 📅 AI Timetable Optimizer

AI — CONSTRAINT OPTIMIZATION

Gemini 2.0 Flash (Constraint Optimization + Text)

Target: Admin Portal — Timetable — Analyzes teacher availability, subject frequency needs, cognitive load sequencing (heavy subjects morning, lighter post-lunch), and classroom conflicts. Auto-generates an optimized conflict-free timetable for Admin to approve in one click.

25. 📄 AI Handwritten Answer Sheet Auto-Grader

AI — VISION

Gemini 2.0 Pro (Vision)

Target: Teacher Portal — Assignments / Classes — Student uploads photo of handwritten descriptive answer. Gemini reads handwriting, compares against teacher's rubric, outputs score and specific feedback on missed key terms. Teacher batch-grades 40 papers in 10 seconds with 1-click approval.

26. 📄 1-Click Textbook PDF → 5-Modality Course Generator

AI — LONG CONTEXT

Gemini 2.0 Pro (2M Token Long Context)

Target: SuperAdmin / Teacher Curriculum Hub — Drop a NCERT CBSE chapter PDF (15–30 pages). Gemini analyzes the whole chapter in one pass and auto-generates: 10 MCQ questions (into question_bank), 5 Story video scripts with timing + narration, Glossary & flashcards, and Virtual Lab experiment specifications. One PDF → complete 5-modality course content.

 COMPLETE PRIORITY ACTION MATRIX (26 RECOMMENDATIONS)

PRIORITY	CATEGORY	RECOMMENDATION	EXPECTED OUTCOME
P0 Urgent	Security	Supabase Row-Level Security (RLS) on all tables	Blocks unauthorized student data reads from browser DevTools
P0 Urgent	Security	Remove ?user= URL param parsing in Chats.jsx	Eliminates stored XSS session hijacking vulnerability
P0 Urgent	Security	Client-side auth → server-side RLS enforcement	Prevents role spoofing and teacher portal unauthorized access
P0 Urgent	Data Integrity	Connect Progress.jsx to real Supabase quiz scores	Students see real academic data, not hardcoded numbers
P0 Urgent	Data Integrity	Persist test scores to test_submissions table	Scores survive device change, browser clear, incognito
P1 High	Architecture	Shared packages/shared/ monorepo	Single source of truth — eliminates 4x code duplication
P1 High	Backend	Create student_enrollments table	Students only see tests and content for their own class
P1 High	UI	Mobile responsive layouts	Platform usable on student phones across India
P1 High	Feature	Student Attendance Tracking System	Completes essential school administrative suite
P1 High	Feature	Gamification & XP System in Study Island	Drives daily student retention, 40–60% engagement lift
P1 High	AI Feature	Snap & Solve Vision Doubt Solver	Instant homework help from camera — 100x faster than typing
P1 High	AI Feature	Canvas-Aware 3D Live Voice Tutor	Personal 1-on-1 physics tutor inside every 3D simulation
P1 High	AI Feature	At-Risk Student Early Warning System	Catches failing students weeks before exams automatically
P1 High	AI Feature	AI Chat Moderation & Academic Integrity	Flags bullying, answer-sharing, and distress signals in real-time
P2 Medium	Architecture	Convert Study Island to modular React app	New chapters = new folders. Zero risk to existing content
P2 Medium	Feature	Homework / Assignment submission system	Full assignment lifecycle: assign → submit → grade
P2 Medium	Feature	Web Push Notifications (native API)	Instant alerts when live class starts or test assigned
P2 Medium	Feature	Parent Portal (read-only)	Parent visibility into attendance, grades, timetable
P2 Medium	AI Feature	CBSE Exam Prediction Engine	Identifies top 5 exam questions + curriculum gaps
P2 Medium	AI Feature	Live Class Transcript & Smart Notes Generator	Auto-notes for every class — benefits absent students too
P2 Medium	AI Feature	Personalized AI Learning Path Engine	Custom chapter sequence per student based on real scores
P2 Medium	AI Feature	Voice-to-Structured MCQ Creator	Teacher speaks — quiz appears. Zero typing required
P2 Medium	AI Feature	Chapter-Scoped RAG Expert Chat	Curriculum-accurate AI answers from chapter content only
P2 Medium	AI Feature	AI Handwritten Answer Sheet Auto-Grader	Batch-grade 40 students in 10 seconds with 1-click
P3 Future	AI Feature	Chapter Storybook Illustrator (Imagen 3)	Rich illustrated stories for history & geography chapters
P3 Future	AI Feature	Vernacular Multilingual Voice Narration	Hindi / Hinglish audio for CBSE accessibility
P3 Future	AI Feature	Dynamic Achievement Badge Generator (Imagen 3)	Personalized shareable milestone badges per student
P3 Future	AI Feature	AI Study Buddy Matchmaking	Pairs complementary students for peer learning
P3 Future	AI Feature	AI Parent Communication Draft Generator	Professional parent letters with real data — 1-click
P3 Future	AI Feature	3D Scene Auto-Generator from Text (Three.js)	New 3D simulations without a 3D developer
P3 Future	AI Feature	AI Timetable Optimizer	Conflict-free optimized schedule generated automatically
P3 Future	AI Feature	1-Click Textbook PDF → 5-Modality Course	Complete course content from a single PDF upload

 **Architecture Note:** All Gemini API calls route through `server.js` backend endpoints — no client-side API key exposure. Native browser APIs (MediaRecorder, WebRTC, canvas.toBlob) used for media capture. Supabase Edge Functions used for real-time moderation. Zero new npm dependencies required for most features.